IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EIE PART II: MEng, BEng and ACGI

Corrected Copy

FEEDBACK SYSTEMS

Thursday, 30 May 10:00 am

Time allowed: 1:30 hours

There are THREE questions on this paper.

Answer ALL questions. Question 1 carries 40% of the marks. Questions 2 and 3 carry equal marks (30% each).

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : F. Teng

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1. Consider the feedback loop shown in Figure 1.1. Here K_p is a constant compensator and $G(s) = G_1(s)G_2(s)$ is the system where each of $G_1(s)$ and $G_2(s)$ is a transfer function representing the circuit shown in Figure 1.2 and where the value of the parameters for $G_1(s)$ are such that $C_i = 0$, $R_iC_f = 1$, $R_f = \infty$, while those for $G_2(s)$ are such that $C_i = 0$, $R_iC_f = 1$, $R_fC_f = 1/2$. Assume all the capacitors are initially uncharged.

a) Determine the transfer function $G(s)$. [5]

b) Write down the differential equation relating $u(t)$ to $y(t)$. [5]

c) Obtain a state–space realisation for $G(s)$. [5]

d) Suppose that $r(t)$ be a unit step applied at $t = 0$. Find the steady–state value of $e(t)$. [5]

e) Let $r(t)$ be a unit step applied at $t = 0$. Find K_p such that the closed–loop response has a time constant of 1 second and a maximum overshoot of 5%. [5]

f) Suppose that $r(t)$ be a unit ramp applied at $t = 0$. For the value of K_p above, use the final value theorem of the Laplace transform, which should be stated, to find the steady–state value of $e(t)$. [5]

g) Draw the Nyquist diagram for $G(s)$. [5]

h) Use the Nyquist criterion to find the number of unstable closed loop poles for all $-\infty < K_p < \infty$. [5]

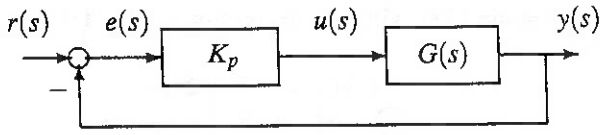


Figure 1.1

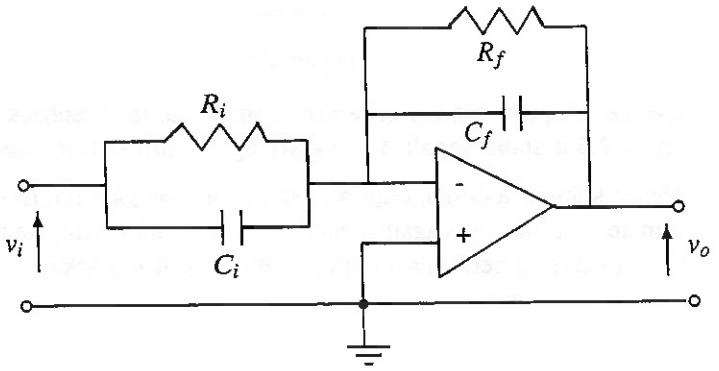


Figure 1.2

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2. In the feedback system in Figure 2.1, $G(s)=\dfrac{2}{(s+1)^3}$ and $K(s)$ is a compensator.

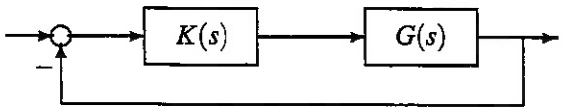


Figure 2.1

a) Sketch the Nyquist diagram of $G(s)$. Use the Routh array to find the real–axis intercepts, together with the corresponding frequencies. [6]

b) Take $K(s)=1$. Use the Routh array to show that the feedback loop is stable. Find the gain and phase margins and the cross–over frequency. [6]

c) Without doing any design, briefly describe how a PI compensator would affect the performance. [6]

d) Suppose that $K(s)$ is a stabilising compensator. Figure 2.2 depicts a stable multiplicative uncertainty $\Delta(s)$.

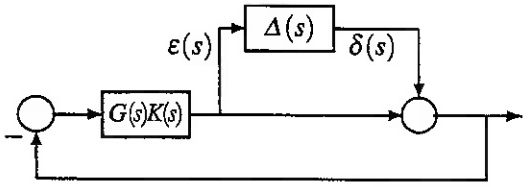


Figure 2.2

i) Show that the feedback loop in Figure 2.2 is equivalent to that in Figure 2.3 for some $C(s)$. Give an expression for $C(s)$ in terms of $G(s)$ and $K(s)$. [4]

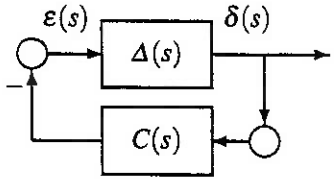


Figure 2.3

ii) Use the Nyquist stability criterion to show that the feedback loop in Figure 2.3 is stable for all $\Delta(s)$ satisfying $|\Delta(j\omega)| < 1/|C(j\omega)|$. [4]

iii) Hence suggest a design requirement on the loop gain $|G(j\omega)K(j\omega)|$ to increase robustness against additive uncertainties. Comment on the effect of this on performance specifications such as tracking. [4]

3. Consider the feedback loop shown in Figure 3.1 below.

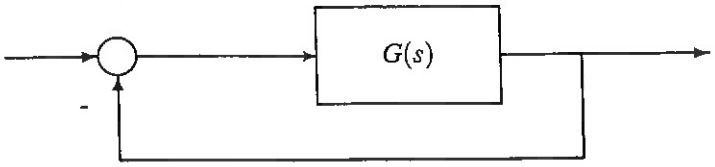


Figure 3.1

a) Suppose that

$$G(s) = \frac{-4s}{s^2 + ks + 4}$$

where $k > 0$ is a design parameter. It is required to find k such that closed–loop response to a step reference signal is critically damped with a settling time of 2 seconds.

i) Find the location of the closed–loop poles that achieves the design specification. [5]

ii) Derive the closed–loop characteristic equation. [5]

iii) Find the value of k that achieves the design specification. [5]

b) Suppose that

$$G(s) = \frac{-4s}{s^2 + k(s - z) + 4}$$

where $k > 0$ and z are design parameters. It is required to find k and z such that closed–loop response to a step reference signal achieves the following design specifications:

- The settling time is at most 2 seconds.
- The response is oscillatory with a maximum overshoot of 5%.

i) Find the location of the closed–loop poles that achieves the design specification. [5]

ii) Derive the closed–loop characteristic equation. [5]

iii) Find the values of k and z that achieve the design specifications. [5]